

Technology Stack & Architecture Matrix

Date	31 January 2025
Team ID	
Project Name	FreelanceHub — Freelancing Marketplace
Maximum Marks	4 Marks

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive decoupled Single Page Application handling client state, job listings, and role-based dashboards.	React.js, JavaScript, HTML5, CSS3, React Router Dom
2	Application Logic-1 (Frontend Broker)	Asynchronous client service layers handling endpoint communications for jobs, proposals, and chat.	Axios HTTP Broker Layer
3	Application Logic-2 (Backend Services)	Stateless middleware handling routes, access filters, and payload processing for projects, proposals, and users.	Node.js, Express.js Framework
4	Application Logic-3 (Security Pipeline)	Specialized identity authentication gatekeepers decoding payload headers and enforcing client/freelancer/admin role access.	JSON Web Token (JWT) Cryptographic Bearer Stack
5	Database	Document mapping storage layer guaranteeing collection data model accuracy across users, projects, and transactions.	MongoDB (NoSQL Document Store)
6	Cloud Database / Persistence Mapping	Object Document Mapper schema enforcement filters for local/remote targets.	Mongoose ODM & MongoDB Atlas connection URI
7	Infrastructure / Server Deployment	Execution runtime engines and execution environments for local sandboxes.	Node.js Runtime Environment v16.x or superior

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology Implementation
1	Open-Source Frameworks	Decoupled web frameworks orchestrating user environments and structural document endpoints for hiring workflows.	MERN Stack ecosystem (MongoDB, Express.js, React, Node.js)
2	Security Implementations	Stateless authorization pipelines encrypting identity data credentials with custom secrets and role-gated route access.	Server-side cryptographically signed JWT strings, Bcrypt hashed schema parameters
3	Scalable Architecture	Three-tier layered separation partitioning visual templates completely from database execution routes.	Decoupled Full-Stack Architecture via Mongoose data persistency modeling
4	Availability	Continuous execution capability across dynamic route templates and environment clusters.	Decoupled asynchronous Node.js server pipelines interacting with MongoDB Atlas clusters
5	Performance	Engineered for sub-second visual transitions and ephemeral memory hook synchronization loops.	React state hooks, local caching storage engines, and non-blocking Axios service broker